FOUR GENERAL FACTS ABOUT COMPLETION: SURJECTIVITY, EXACTNESS FOR FLAT QUOTIENTS, AND FINITENESS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Nakayama applied over a nilpotent thickening. Invoked in the counted proof at lines 22805-22808. Lifts surjectivity modulo a power of the ideal to surjectivity of the system of quotients.
- Left exactness of inverse limits. Used in the counted proof (the exactness statement). Gives exactness of the completed sequence except at the last term, which flatness supplies.

Source reference. Original lemma 03CA: sources/stacks-algebra/pinned-original-source.tex, source lines 21065–21065. Navigation pointer for \ref{lemma-Mittag-Leffler} used inside the counted span (line 22823); the argument that consumes it is retained in the span.

Source reference. Original lemma 00DV: sources/stacks-algebra/pinned-original-source.tex, source lines 3833–3833. Navigation pointer for \ref{lemma-NAK} used inside the counted span (line 22806); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00HL: sources/stacks-algebra/pinned-original-source.tex, source lines 9254–9254. Navigation pointer for \ref{lemma-flat-tor-zero} used inside the counted span (line 22831); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $I \subset R$ be an ideal. Let $\varphi : M \to N$ be a map of R-modules.*

1. *If $M/IM \to N/IN$ is surjective, then $M^\wedge \to N^\wedge$ is surjective.*
2. *If $M \to N$ is surjective, then $M^\wedge \to N^\wedge$ is surjective.*
3. *If $0 \to K \to M \to N \to 0$ is a short exact sequence of R-modules and N is flat, then $0 \to K^\wedge \to M^\wedge \to N^\wedge \to 0$ is a short exact sequence.*
4. *The map $M \otimes_R R^\wedge \to M^\wedge$ is surjective for any finite R-module M.*

Proof. Assume $M/IM \to N/IN$ is surjective. Then the map $M/I^nM \to N/I^nN$ is surjective for each $n \geq 1$ by Nakayama's lemma. More precisely, apply Lemma 00DV part (11) to the map $M/I^nM \to N/I^nN$ over the ring R/I^n and the nilpotent ideal I/I^n to see this. Set $K_n = \{x \in M \mid \varphi(x) \in I^nN\}$. Thus we get short exact sequences

$$0 \to K_n/I^nM \to M/I^nM \to N/I^nN \to 0$$

We claim that the canonical map $K_{n+1}/I^{n+1}M \to K_n/I^nM$ is surjective. Namely, if $x \in K_n$ write $\varphi(x) = \sum z_j n_j$ with $z_j \in I^n$, $n_j \in N$. By assumption we can write $n_j = \varphi(m_j) + \sum z_{jk} n_{jk}$ with $m_j \in M$, $z_{jk} \in I$ and $n_{jk} \in N$. Hence

$$\varphi(x - \sum z_j m_j) = \sum z_j z_{jk} n_{jk}.$$

This means that $x' = x - \sum z_j m_j \in K_{n+1}$ maps to $x \bmod I^nM$ which proves the claim. Now we may apply Lemma 03CA to the inverse system of short exact sequences above to see (1). Part (2) is a special case of (1). If the assumptions of (3) hold, then for each n the sequence

$$0 \to K/I^nK \to M/I^nM \to N/I^nN \to 0$$

is short exact by Lemma 00HL. Hence we can directly apply Lemma 03CA to conclude (3) is true. To see (4) choose generators $x_i \in M$, $i = 1, \ldots, n$. Then the map $R^{\oplus n} \to M$, $(a_1, \ldots, a_n) \mapsto \sum a_i x_i$ is surjective. Hence by (2) we see $(R^\wedge)^{\oplus n} \to M^\wedge$, $(a_1, \ldots, a_n) \mapsto \sum a_i x_i$ is surjective. Assertion (4) follows from this. □